

MySQL

- * MySQL କଣ; Relational DataBase Management System ଥିବାକି। (RDBMS)
- * RDBMS ଥିବାର କଣ; relational database ଥିବାର maintain କରିବା କମ୍ପ୍ୟୁଟରୀୟ programme ଥିବାକି।
- * RDBMS ରଖିବା database ପରିଷ୍କାରିବା data access କରିବା 'SQL queries' ଦେବାକି।
- * Relational database ଥିବାର କଣ; data tables ପ୍ରତି ପ୍ରତି tables ମଧ୍ୟରେ ଯୋଗ ସଂଯୋଜନା କରିବା ଏବଂ data ହାତକୁଣ୍ଡଳୀ ବାହାରେ, ଥିବା data tables ମଧ୍ୟରେ relationships ସୃଷ୍ଟି କରିବା database ଥିବାକି।
- * MySQL ପ୍ରତି କେଉଁ କେଉଁ SQL statement ଥିବାର ଅନୁସନ୍ଧାନ କରିବା semicolon (;) ଦେବାକି।

SELECT

- * database থেকে data select করার SELECT statement রয়েছে।
- * The data returned is stored in a result table, called the result-set.
- * এর ধারণা হল syntax হলো হলো।

* columns ଅବିଚୟକା' ଚାହନ୍ଦା' select ଅବିଚୟକା;
SELECT column1, column2, ...
FROM table_name;

* ପୂର୍ଣ୍ଣ table ଡାଟା select କରିବେ;
SELECT * FROM table-name;

200- 1.) `SELECT CustomerName, City, Country FROM Customers;`

2.) SELECT * FROM Customers;

SELECT DISTINCT

* ଉଦାହରଣ 'Customers' demo database
 ନିମ୍ନ ଗଠନ ଥାଏ।

- ★ ડિસ્ટિનક્ટ વેલ્યુ (distinct / different) values જોઈએ select ઓર્ડર બદલાઈ.
- ★ વેલ્યુ syntax થી જાણ કરી શકો.

```
SELECT DISTINCT column1, column2, ...  
FROM table_name;
```

```
2207 SELECT DISTINCT Country FROM Customers;
```

- (*) COUNT statement দেয় একাধিক distinct values এর সংখ্যা জানায়।

```
2204 SELECT COUNT (DISTINCT Country) FROM Customers;
```

WHERE

- * කිසියම් විශේෂිත කොන්දේසිකර් තවදුරටත් records එමගින් extract කිරීමේ මෙය සිදුකරයි.
- * ඔබේ syntax එය අවිනි වේ.

```
SELECT column1, column2, ...  
FROM table-name  
WHERE condition;
```

* WHERE clause එක SELECT හි දී එමගින් ලබාදේ; UPDATE, DELETE ප්‍රතිපත්ති සඳහා වේ.

උදා:- `SELECT * FROM Customers
WHERE Country = 'Mexico';`

- * WHERE clause එක අඩුම ඔබේ operators සඳහා වේ.

Operator	Description
=	Equal
>	Greater than
<	Less than
>=	Greater than or Equal
<=	Less than or Equal
<>	Not equal. (Some versions of SQL this operator written !=)
BETWEEN	Between a certain range
LIKE	Search for a pattern
IN	To specify multiple possible values for a column

AND, OR, NOT

- * WHERE clause එක AND, OR හෝ NOT operators ඔබේ ද, සඳහා වේ.

- * condition සියලුම true වේ; කිසියම් record එකක් display කිරීමට AND ද,
- * conditions ඉතිරි එකක් හෝ true වේ; කිසියම් record එකක් display කිරීමට OR ද,
- * condition එක false වේ; කිසියම් record එකක් display කිරීමට NOT ද,

සොයා ගන්න.

★ AND syntaxes જોવા સર્જે છે.

★ AND syntax ;

```
SELECT column1, column2, ...  
FROM table-name  
WHERE condition1 AND condition2 ... ;
```

★ OR syntax ;

```
SELECT Column1, column2, ...  
FROM table-name  
WHERE condition1 OR condition2 ... ;
```

★ NOT syntax ;

```
SELECT column1, column2, ...  
FROM table-name  
WHERE NOT condition ;
```

PRO :- 1.) SELECT * FROM Customers

WHERE Country = 'Germany' AND City = 'Berlin' ;

2.) SELECT * FROM Customers

WHERE City = 'Berlin' OR City = 'Stuttgart' ;

3.) SELECT * FROM Customers

WHERE NOT Country = 'Germany' ;

★ AND, OR અને NOT combined કરીને શ્રેણી, statements લેઈ શકાય.

ORDER BY

★ result set યોજનાં ascending અને descending order યોજાવે છે. આથી તેને સરળ બનાવે છે.

★ Default યોજનાં ascending order યોજાવે છે; descending order એવું જણાવવા માટે DESC keyword યોજાવે છે.

★ Syntax યોજાવે છે.

```
SELECT column1, column2, ...  
FROM table-name  
ORDER BY column1, column2, ... ASC | DESC ;
```

୧୧୦୮ SELECT * FROM Customers (* Ascending order)
ORDER BY Country;

୧୧୦୮ SELECT * FROM Customers
ORDER BY Country DESC;

⊛ columns ରେ ଗଠନକାରୀ ହେଉ ୧, କେହିକିଛି ଧାରଣକାରୀ ନୁହେଁ.

୧୧୦୮ SELECT * FROM Customers
ORDER BY Country, CustomerName;

୧୧୦୮ SELECT * FROM Customers
ORDER BY Country ASC, CustomerName DESC;

INSERT INTO

⊛ table ଥିବାକାରୀ ସ୍ତମ୍ଭର 'records' ଥିବା ସ୍ଥିତିରେ କେତେକ.

⊛ INSERT INTO statements ସାହାଯ୍ୟରେ ଡେଟାବାସୀ ଲେଖି ଥାଆନ୍ତି. Syntaxes ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି.

1.) INSERT INTO table-name (column1, column2, column3,...)
VALUES (value1, value2, value3,...);

* ଯେଉଁଠି columns ରେ ଗୋଟିଏ ସ୍ଥାନରେ ଥିବା, ଗୋଟିଏ କେତେକ ସ୍ଥିତିରେ values ଗ୍ରହଣକାରୀ ହେଉ ଥାଆନ୍ତି.

* ସ୍ତମ୍ଭର ଗୋଟିଏ ସ୍ଥିତିରେ columns ରେ ଥିବା ସ୍ଥାନରେ ସ୍ତମ୍ଭର columns ରେ ଥିବା ସ୍ଥାନରେ values କେତେକ ଥାଆନ୍ତି.

* ଯଦି, ଗୋଟିଏ values 'NULL' ରେଖି ଥିବା ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି.

2.) INSERT INTO table-name
VALUES (value1, value2, value3,...);

* ଯେଉଁଠି column names ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି; columns ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି values କେତେକ ଥାଆନ୍ତି.

NULL values

⊛ value ଥିବାକାରୀ ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି field ଥିବାକାରୀ, NULL value ସ୍ଥାନରେ ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି field ଥିବାକାରୀ.

⊛ NULL value ଥିବାକାରୀ ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି; zero (0) value ଥିବା ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି; whitespaces ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି value ଥିବାକାରୀ ୧ ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି.

⊛ NULL values ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି IS NULL ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି IS NOT NULL operators ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି.

୧୧୦୮ 1.) SELECT CustomerName, Address
FROM Customers
WHERE Address IS NULL;

2.) SELECT CustomerName, Address
FROM Customers
WHERE Address IS NOT NULL;

UPDATE

- ⊛ table එකේ record එකක් update කිරීමට යොදා.
- ⊛ Syntax එක පහත පරිදි වේ.

UPDATE table-name

SET column 1 = value 1, column 2 = value 2, ...

WHERE condition;

Note: records update කිරීමේ දී ඉහත කාලිසංකීර්ති පිට ඇති, මෙහි WHERE clause එක වෙනස්වන්නේ කවුරුත් table එකම update වේ!

උදා - UPDATE Customers;

SET ContactName = 'Alfred', City = 'Frankfurt'

WHERE CustomerID = 1;

උදා - UPDATE Customers

SET PostalCode = 00000

WHERE Country = 'Mexico';

DELETE

- ⊛ table එකෙහි records delete කිරීමට යොදා.
- ⊛ Syntax එක පහත පරිදි වේ.

DELETE FROM table-name WHERE condition;

Note: records delete කිරීමේ දී ඉහත කාලිසංකීර්ති පිට ඇති, මෙහි WHERE clause එක වෙනස්වන්නේ කවුරුත් table එකම delete වේ!

උදා 1) DELETE FROM Customers

WHERE CustomerName = 'Alfreds';

2) සියලු records delete කිරීමට,

(DELETE FROM table-name;) Syntax යොදා.

DELETE FROM Customers;

LIMIT

- ⊛ return කළ යුතු records ගණන දැක්වීමට LIMIT clause යොදා.
- ⊛ පහත Syntax එක යොදා.

```
SELECT column-name(s)
FROM table-name
WHERE condition
LIMIT number;
```

ଉଦା:- SELECT * FROM Customers
LIMIT 3;

* 4-6 ମଧ୍ୟ records ଡାଟାବେସ୍‌ରୁ ଫିରସ୍ତ କରିବା ପାଇଁ ଏହା କିଛି ସମୟ ଲାଗିବ।
OFFSET ବ୍ୟବହାର କରନ୍ତି।

ଉଦା:- SELECT * FROM Customers
LIMIT 3 OFFSET 3;

* This will return only 3 records,
start on record 4 (OFFSET 3)

④ ଡାଟାବେସ୍‌ରୁ ଡାଟା ଫିରସ୍ତ କରିବା ପାଇଁ statements ଉପରେ WHERE clause ଥିବା ୧, କେହି ନାହିଁ।

MIN(), MAX()

④ ଡାଟାବେସ୍‌ରୁ column ଥିବା ସ୍ତମ୍ଭରୁ ସର୍ବନିମ୍ନ ଓ ସର୍ବୋଚ୍ଚ values ଖୋଜିବା ପାଇଁ MIN() ଓ MAX() on-build functions ବ୍ୟବହାର କରନ୍ତି।

MIN() Syntax

```
SELECT MIN(column-name)
FROM table-name
WHERE condition;
```

MAX() Syntax

```
SELECT MAX(column-name)
FROM table-name
WHERE condition;
```

ଉଦା:- SELECT MIN(Price) AS SmallestPrice
FROM Products;

ଉଦା:- SELECT MAX(Price)
AS LargestPrice
FROM Products;

COUNT(), AVG(), SUM()

④ ଡାଟାବେସ୍‌ରୁ ଡାଟା ଫିରସ୍ତ କରିବା ପାଇଁ columns ଉପରେ ~~କେହି ନାହିଁ~~ ଡାଟାବେସ୍‌ରୁ COUNT() on-build function ବ୍ୟବହାର କରନ୍ତି।

④ Syntax ଥିବା ଅନୁସାରେ ଡାଟା ଫିରସ୍ତ କରିବା ପାଇଁ।

```
SELECT COUNT(column-name)
FROM table-name
WHERE condition;
```

* WHERE clause ବ୍ୟବହାର କରିବା ପାଇଁ, ଡାଟା ଫିରସ୍ତ କରିବା ପାଇଁ।

ଉଦା:- SELECT COUNT(ProductID)
FROM Products;

Note: NULL values are not counted

- numeric column થી average થી બાહીર AVG() on-build function બોલે.
- ★ Syntax થી રજા કરે છે.

```
SELECT AVG (column-name)
FROM table-name
WHERE condition;
```

★ WHERE clause જોડાઈ શકે છે, બેકેક થાય.

જેમ-
SELECT AVG (Price)
FROM Products;

Note: NULL values are ignored

- ★ numeric column થી સરેરાશ થી બાહીર SUM() on-build function બોલે.
- ★ Syntax રજા કરે છે.

```
SELECT SUM (column-name)
FROM table-name
WHERE condition;
```

★ WHERE clause જોડાઈ શકે છે, બેકેક થાય.

જેમ-
SELECT SUM (Quantity)
FROM OrderDetails;

Note: NULL values are ignored

LIKE

- ★ column થી જુદા પેટર્ન થી બાહીર WHERE clause થી રજા કરે છે.
- ★ LIKE operator થી જોડાઈ શકે છે અને 'wildcards' રજા કરે.

- The percent sign (%) represents zero, one or multiple characters
- The underscore sign (_) represents one, single character

- ★ Syntax થી રજા કરે છે.

```
SELECT column1, column2, ...
FROM table-name
WHERE columnN LIKE pattern;
```

- ★ '%' અને '_' જોડાઈ શકે છે LIKE operators થી રજા કરે છે.

LIKE operator	Description
WHERE CustomerName LIKE 'a%'	Finds any values starts with "a"
WHERE CustomerName LIKE '%a'	Finds any values ends with "a"
WHERE CustomerName LIKE '%or%'	Finds any values have "or" any position
WHERE CustomerName LIKE '_r%'	Finds any values have "r" second position
WHERE CustomerName LIKE 'a%o'	Finds any values starts with "a" ends with "o"

ଉଦା:- 1) SELECT * FROM Customers

WHERE CustomerName LIKE 'a_';

IN

- ① values ବଦଳାଇବା ବିକଳ୍ପିତା ଥିବାବେଳେ WHERE clause ଥିବା **IN** operator ବ୍ୟବହାର କରାଯାଏ.
- ② **IN** operator, multiple OR conditions ଯେଉଁଠି shorthand ଢଙ୍ଗରେ ଲେଖିବା ସମ୍ଭବ.
- ③ Syntax ଥିବା ଅନୁସାରେ ଗ୍ରହଣ କରାଯାଏ.

SELECT column-name(s)

FROM table-name

WHERE column-name IN (value1, value2, ...);

ଉଦା:-

SELECT column-name(s)

FROM table-name

WHERE column-name IN (SELECT STATEMENT);

- ④ ଅନୁସାରେ ଉଦାହରଣ ଦିଆଯାଉଛି.

ଉଦା:- 1) SELECT * FROM Customers

WHERE Country IN ('Germany', 'France', 'UK');

2) SELECT * FROM Customers

WHERE Country NOT IN ('Germany', 'France', 'UK');

3) SELECT * FROM Customers

WHERE Country IN (SELECT Country FROM Suppliers);

BETWEEN

- ① range ବଦଳାଇବା 2 values select କରିବାର ବ୍ୟବହାର.

- ② ଅନୁସାରେ Syntax ଗ୍ରହଣ କରାଯାଏ.

SELECT column-name(s)

FROM table-name

WHERE column-name BETWEEN value1 AND value2;

ଉଦା:- SELECT * FROM Products

WHERE Price BETWEEN 10 AND 20;

- ③ range ବଦଳାଇବା ବିକଳ୍ପିତା 2 values select କରିବାର NOT BETWEEN ବ୍ୟବହାର କରାଯାଏ.

ଉଦା:- SELECT * FROM Products

WHERE Price NOT BETWEEN 10 AND 20;

numeric values સંખ્યાત્મક છે, text values અક્ષર ૨ લેખ લેખિત થાય.

Aliases

- ★ table ଥିବାକାର ଡେଇଁ table column ଥିବାକାର temporary name ଥିବାକାର ଫିରର ଚିହ୍ନ ଲେଖାଯାଇଛି.
- ★ ଚିହ୍ନ ଡାହାଣ କରାଯାଇଛି; column names more readable କରିବାକୁ ଅନୁମତି.
- ★ AS keyword ଡାହାଣ ଲେଖାଯାଇଥିବା ମୁଖ୍ୟ, ଲେଖାଯାଇଥିବାକାର ଥିବା query ଥିବାକାର duration ଥିବା ଅନୁମତି.
- ★ Syntaxes ଡାହାଣ କରିବା ଚିହ୍ନ.

Column Syntax

```
SELECT column_name AS alias_name  
FROM table_name;
```

```
(QOT 1) SELECT CustomerID AS ID,  
CustomerName AS Customer  
FROM Customers;
```

```
2) SELECT CustomerName AS Customer,  
    ContactName AS "Contact Person"  
    FROM Customers;
```

- Alias name should spaces গণনা করা হবে না, single / double quotation marks দিয়ে লিখতে হবে।

Table Syntax

```
SELECT column_name(s)
FROM table_name AS alias_name;
```

```

201-1) SELECT o.OrderID, o.OrderDate,
        c.CustomerName
FROM Customers AS c,
Orders AS o
WHERE c.CustomerName = X
AND c.CustomerID = o.CustomerID;

```

JOIN

- ⊛ tables ເຂົາເຈົ້າ ຂອງ ວິທີການທີ່ ການ related column ທີ່ມີ ຈຳນວນ, ~~rows~~ rows combine ສົມທົບ ເອກະລັກ.
- ⊛ "Orders" ແລະ "Customers" ໃນ demo databases (tables) ເພື່ອໃຫ້ 'Customer ID' column ທີ່ມີ ງ່າຍດາຍ ເປັນ.
- ⊛ ຫຼັງ ຈາກການ INNER JOIN statement ທີ່ມີ ເອກະລັກ tables ເພື່ອໃຫ້ matching values select ສົມທົບ ດຽວ ເປັນ.

```
2201 SELECT Orders.OrderID, Customers.CustomerName, Orders.OrderDate
      FROM Orders
      INNER JOIN Customers ON Orders.CustomerID = Customers.CustomerID;
```

④ MySQL support କାର୍ଯ୍ୟ join types 4 ଅଟେ.

* INNER JOIN $\Rightarrow$ Returns records that have matching values in both tables

* LEFT JOIN $\Rightarrow$ Returns all records from the left table and the matched records from the right table

* RIGHT JOIN $\Rightarrow$ Returns all records from the right table and the matched records from the left table

* CROSS JOIN $\Rightarrow$ Returns all records from both tables

④ INNER JOIN, LEFT JOIN ଓ RIGHT JOIN ପାଇଁ ଏବଂ CROSS JOIN ନି syntax ଥିବା ଅବଦାନ ଦିଆଯାଇଛି.

```
SELECT column-name(s)
FROM table1
CROSS JOIN table2;
```

```
SELECT Customers.CustomerName, Orders.OrderID
FROM Customers
CROSS JOIN Orders;
```

UNION

④ SELECT statements ଦ୍ଵାରା ଗଠିତ ଉଦ୍ଦିଷ୍ଟ ଫଳାଫଳ result-sets combine କରିବା ଦେଖିବାକୁ ମିଳେ.

④ * Every SELECT statement within UNION must have the same number of columns

* The columns must also have similar data types

* The columns in every SELECT statement must also be in the same order

④ ଏହାର syntax ଥିବା ଅବଦାନ ଦିଆଯାଇଛି.

```
SELECT column-name(s) FROM table1
UNION
SELECT column-name(s) FROM table2;
```

* UNION operator ଥିବା default ହେତୁ ପ୍ରଦର୍ଶିତ distinct values select କରିବା ସମ୍ଭବ.

* duplicate values ସମ୍ଭବ ହେତୁ, select କରିବା ସମୟରେ ଏହା UNION ALL operator ଥିବା ବ୍ୟବହାର କରାଯାଏ.

```
SELECT column_name(s) FROM table1
UNION ALL
SELECT column_name(s) FROM table2;
```

ଉଦା: 1) SELECT City FROM Customers
UNION
SELECT City FROM Suppliers
ORDER BY City;

2) SELECT City FROM Customers
UNION ALL
SELECT City FROM Suppliers
ORDER BY City;

GROUP BY

① ଯଦି values ଗ୍ରାହକଙ୍କର rows group କରା ଯାଏ ତେବେ summary rows ଭାବେ ଉଦାହରଣ ଦିଅନ୍ତୁ।

② Syntax ଯାହା ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି।

```
SELECT column_name(s)
FROM table-name
WHERE condition
GROUP BY column_name(s)
ORDER BY column_name(s)
```

ଉଦା: 1) SELECT COUNT(CustomerID), Country
FROM Customers
GROUP BY Country;

2) SELECT COUNT(CustomerID), Country
FROM Customers
GROUP BY Country
ORDER BY COUNT(CustomerID) DESC;

HAVING

① HAVING clause ଯଦି ଉପରେ, aggregate functions ଉପରେ WHERE keyword ଭଳି କାର୍ଯ୍ୟକାରୀ କରାଯାଏ।

② Syntax ଯାହା ଉପରେ ଉଲ୍ଲେଖ କରାଯାଇଛି।

```

SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
HAVING condition
ORDER BY column_name(s);

```

૨૨૦૧- SELECT COUNT (CustomerID), Country
FROM Customers
GROUP BY Country
HAVING COUNT (CustomerID) > 5;

૨૨૦૨- SELECT Employees.LastName, COUNT (Orders.OrderID) AS NumberOfOrders
FROM (Orders
INNER JOIN Employees ON Orders.EmployeeID = Employees.EmployeeID)
GROUP BY LastName
HAVING COUNT (Orders.OrderID) > 10;

EXISTS

- ★ એક ક્વેરીમાં, જ્યાં record છે તે subquery છે તે exist થી તપાસવામાં આવે છે. (used to test for the existence of any record in a subquery)
- ★ EXIST operator એ TRUE લેતો return કરે છે, subquery એમાં એક records હોવાથી return કરે છે.
- ★ Syntax એક છે તેવું.

```

SELECT column_name(s)
FROM table_name
WHERE EXISTS
(SELECT column_name FROM table_name WHERE condition);

```

૨૨૦૩- SELECT SupplierName
FROM Suppliers
WHERE EXISTS
(SELECT ProductName FROM Products WHERE Products.SupplierID =
~~SupplierID~~ Suppliers.SupplierID AND Price < 20);

ANY and ALL

★ Single column value থাকার এক সার values range থাকার সহ comparison থাকার কাজের এক operators ব্যবহার.

★ ANY operator:

* returns a boolean value as a result

* returns TRUE if any of the subquery values meet the condition

ANY means that the condition will be true if the operation is true for any of the values in the range.

★ ALL operator:

* returns a boolean value as a result

* returns TRUE if all of the subquery values meet the condition

* is used with SELECT, WHERE and HAVING statements

ALL means that the condition will be TRUE only if the operation is true for all values in the range.

INSERT INTO SELECT

★ এক table থাকার data copy কর, এক table থাকার insert কাজের ব্যবহার.

★ এক কাজ; source থাকার এক target table থাকার data types এক কাজ কর.

★ Syntaxes এক কাজ কর.

Copy all columns from one table to another

```
INSERT INTO table 2  
SELECT * FROM table 1  
WHERE condition;
```

Copy only some columns from one to another

```
INSERT INTO table 2 (column1, column2,  
SELECT column1, column2, ...  
FROM table 1  
WHERE condition;
```

১) INSERT INTO Customers (CustomerName, City, Country)
SELECT SupplierName, City, Country FROM Suppliers;

২) INSERT INTO Customers (CustomerName, ContactName, Address, City, PostalCode, Country)
SELECT SupplierName, ContactName, Address, City, PostalCode, Country
FROM Suppliers;

৩) INSERT INTO Customers (CustomerName, City, Country)
SELECT SupplierName, City, Country FROM Suppliers
WHERE Country = 'Germany';

CASE

- ① લેક if-else પ્રમાણમાં સુલભ છે.
- ② CASE statement એકાદ છે conditions ઠીકાં કરી શકે; true એ કદી condition એકાદ છે value એકાદ return કરે.
- ③ એકાદ; condition એકાદ true છે; CASE statement એકાદ ઉકારે read કરી શકે.
- ④ કદી condition એકાદ true એકાદ છે; ELSE એકાદ કદી કરે; ELSE એકાદ એકાદ છે; NULL એક return કરે.
- ⑤ Syntax કદી ઠીક છે.

CASE

```
WHEN condition1 THEN result1  
WHEN condition2 THEN result2  
WHEN condition3 THEN result3  
ELSE result
```

END;

જેમક SELECT OrderID, Quantity,
CASE

```
WHEN Quantity > 30 THEN 'The quantity is greater than 30'  
WHEN Quantity = 30 THEN 'The quantity is 30'  
ELSE 'The quantity is under 30'
```

END AS QuantityText

FROM OrderDetails;

જેમક SELECT CustomerName, City, Country

FROM Customers

ORDER BY

(CASE

```
WHEN City IS NULL THEN Country
```

```
ELSE City
```

END));

IFNULL() and COALESCE()

- ★ જો કોઈ value છે તો NULL નથી; અથવા કોઈ value છે તો return કરે જાણીતો IFNULL() on-build function છે.

જેમ- SELECT ProductName, UnitPrice * (UnitsInStock + IFNULL
(UnitsOnOrder, 0))
FROM Products;

* જો NULL values આપે તો zero (0) return કરે.

- ★ જો કોઈ કંઈ, COALESCE() on-build function છે.

જેમ- SELECT ProductName, UnitPrice * (UnitsInStock +
COALESCE (UnitsOnOrder, 0))
FROM Products;

Comments

- ★ MySQL માં single line comments જેથી લેખકો (--) ડાઉન કરી શકે.
- ★ multiline comments /* ... */ લેખકો.

જેમ- -- This is single line comment
/* This is
multiline
comment */

Operators

- ★ MySQL માં operators એક જાણીતી વસ્તુ છે.
- ★ Arithmetic operators બધા; +, -, *, /, % આપે છે.
- ★ Bitwise operators;

& - Bitwise AND

| - Bitwise OR

^ - Bitwise exclusive OR

- ★ Comparison operators આપે;

= → equal to

> → greater than

< → less than

>= → greater than or equal to

<= → less than or equal to

<> → not equal to

⊛ Compound operators ;

$+=$ → Add equals

$-=$ → Subtract equals

$*=$ → Multiply equals

$/=$ → Divide equals

$\%=$ → Modulo equals

$\&=$ → Bitwise AND equals

$\wedge=$ → Bitwise exclusive equals

$|*=$ → Bitwise OR equals

⊛ Logical operators ;

Operator	Description
ALL	TRUE if all of the subquery values meet condition
AND	TRUE if all the conditions separated by AND is TRUE
ANY	TRUE if any of the subquery values meet the condition
BETWEEN	TRUE if the operand is within the range of comparisons
EXISTS	TRUE if the subquery returns one or more records
IN	TRUE if the operand is equal to one of a list of expressions
LIKE	TRUE if the operand matches a pattern
NOT	Displays a record if the condition(s) is NOT TRUE
OR	TRUE if any of the conditions separated by OR is TRUE
SOME	TRUE if any of the subquery values meet the condition